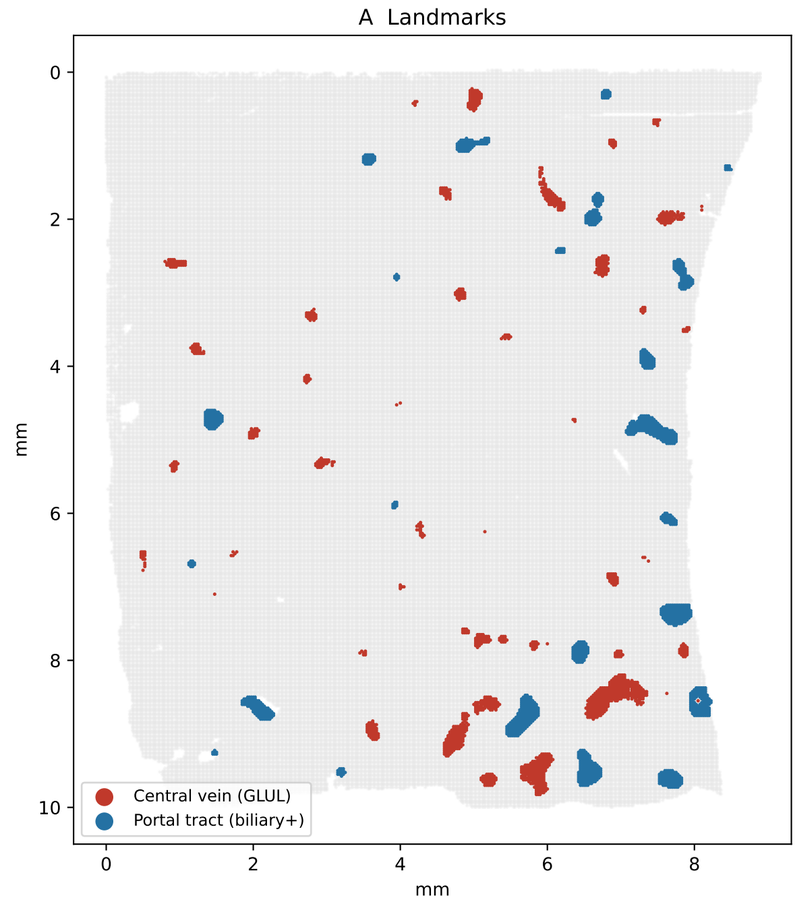
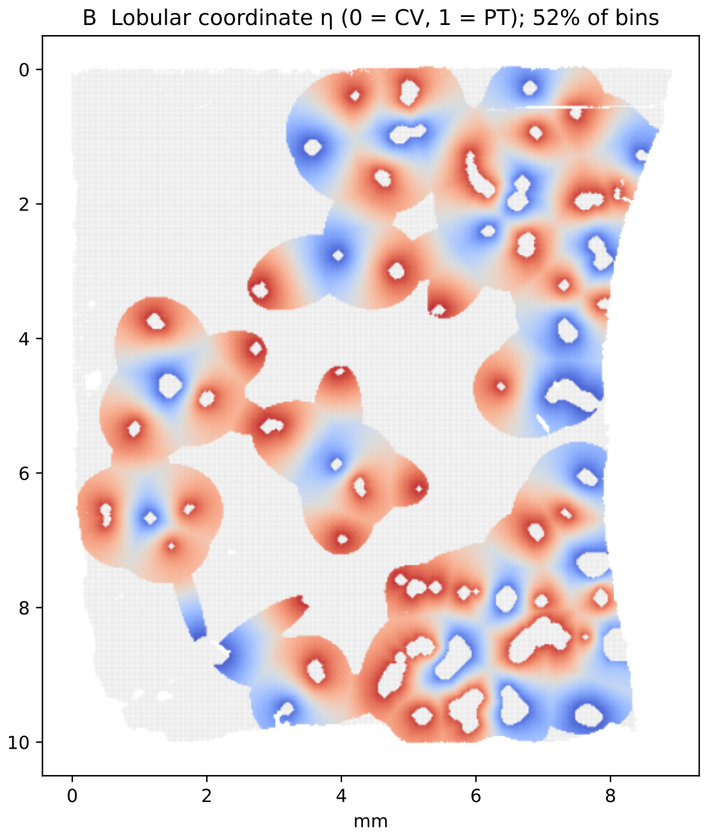
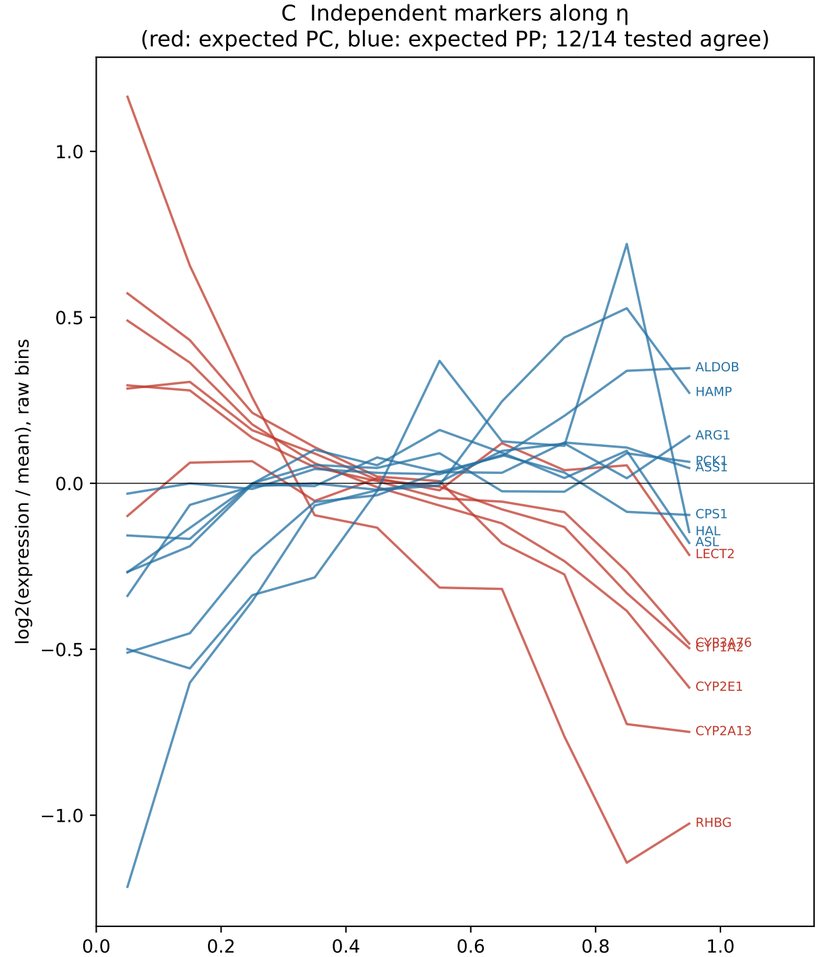
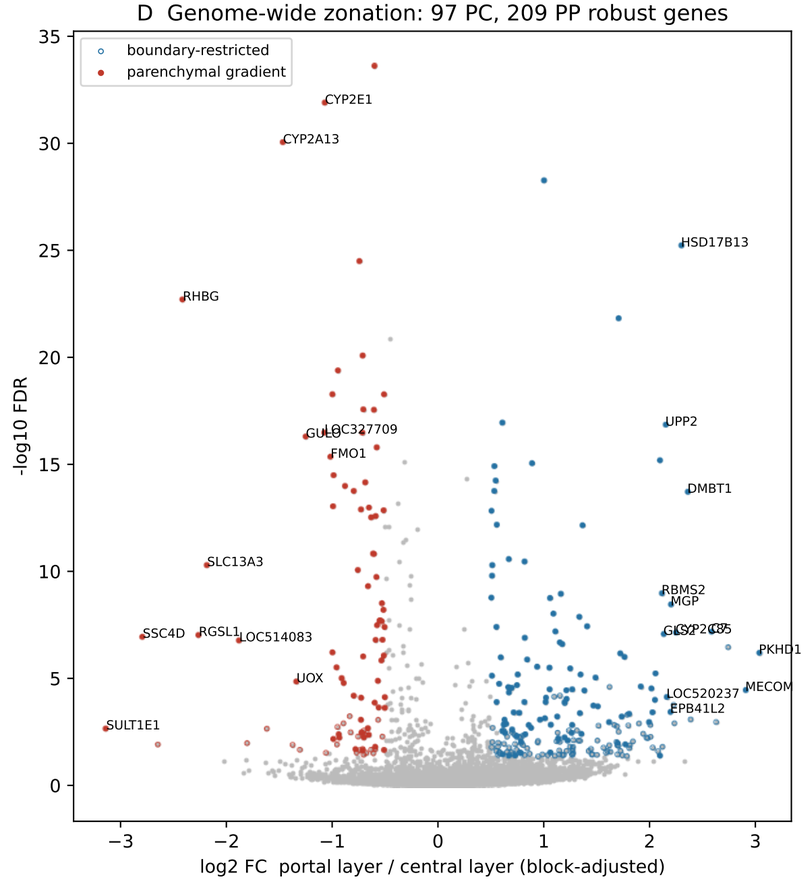
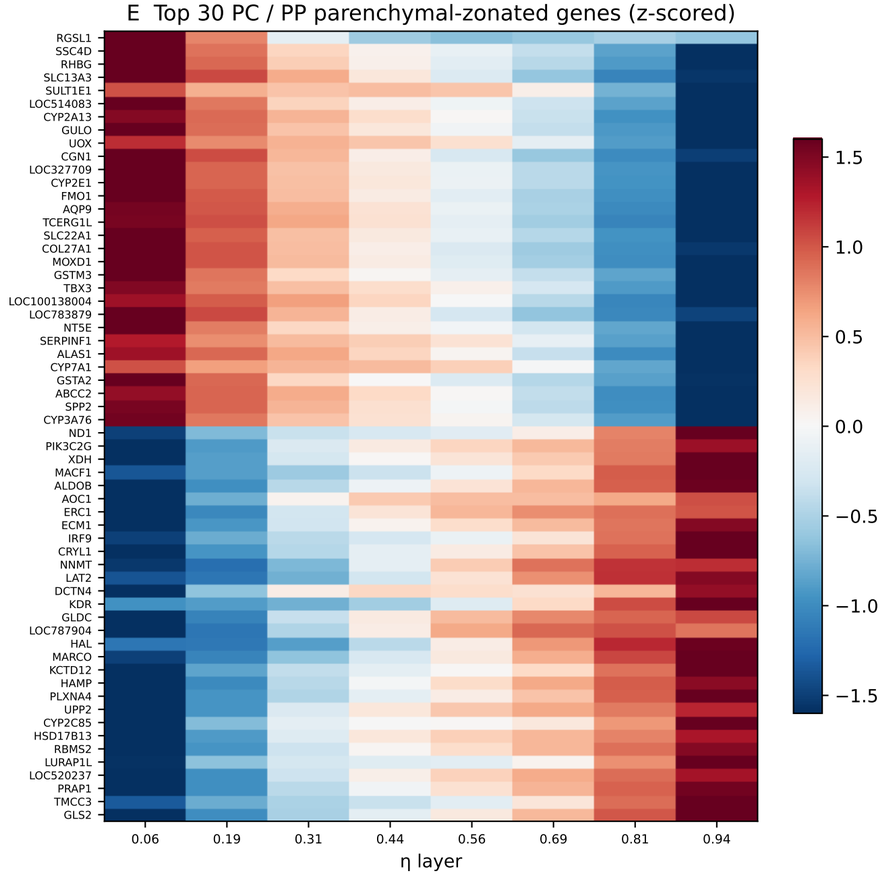
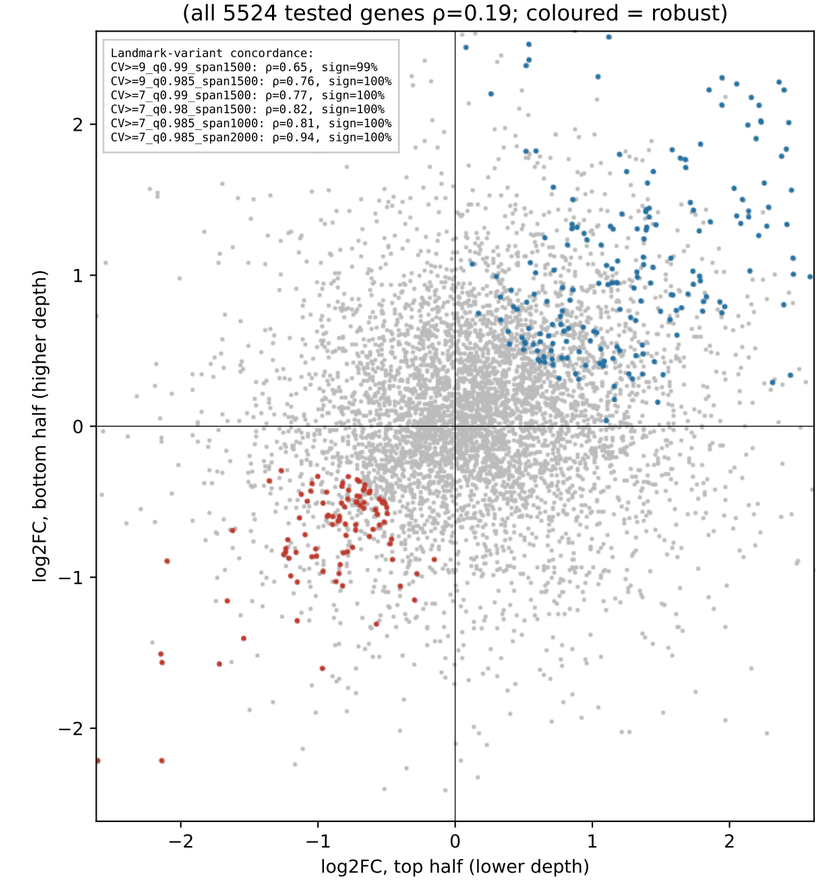

A Landmarks
Central vein (GLUL)
Portal tract (biliary+)
mm
mm


B Lobular coordinate η (0 = CV, 1 = PT); 52% of bins
mm
η


C Independent markers along η
(red: expected PC, blue: expected PP; 12/14 tested agree)
log2(expression / mean), raw bins
η (central → portal)
ALDOB
HAMP
ARG1
PCK1
ASS1
CPS1
HAL
ASL
LECT2
CYP3A76
CYP1A2
CYP2E1
CYP2A13
RHBG


D Genome-wide zonation: 97 PC, 209 PP robust genes
boundary-restricted
parenchymal gradient
-log10 FDR
log2 FC portal layer / central layer (block-adjusted)
CYP2E1
CYP2A13
HSD17B13
RHBG
LOC327709
GULO
FMO1
UPP2
DMBT1
SLC13A3
RBMS2
MGP
SSC4D
RGSL1
LOC514083
PKHD1
UOX
LOC520237
MECOM
EPB41L2
SULT1E1


E Top 30 PC / PP parenchymal-zonated genes (z-scored)
RGSL1
SSC4D
RHBG
SLC13A3
SULT1E1
LOC514083
CYP2A13
GULO
UOX
CGN1
LOC327709
CYP2E1
FMO1
AQP9
TCERG1L
SLC22A1
COL27A1
MOXD1
GSTM3
TBX3
LOC100138004
LOC783879
NT5E
SERPINF1
ALAS1
CYP7A1
GSTA2
ABCC2
SPP2
CYP3A76
ND1
PIK3C2G
XDH
MACF1
ALDOB
AOC1
ERC1
ECM1
IRF9
CRYL1
NNMT
LAT2
DCTN4
KDR
GLDC
LOC787904
HAL
MARCO
KCTD12
HAMP
PLXNA4
UPP2
CYP2C85
HSD17B13
RBMS2
LURAP1L
LOC520237
PRAP1
TMCC3
GLS2
0.06
0.19
0.31
0.44
0.56
0.69
0.81
0.94
η layer


F Split-half replication: zonated genes ρ=0.76, same sign 99%
(all 5524 tested genes ρ=0.19; coloured = robust)
Landmark-variant concordance:
CV>=9_q0.99_span1500: ρ=0.65, sign=99%
CV>=9_q0.985_span1500: ρ=0.76, sign=100%
CV>=7_q0.99_span1500: ρ=0.77, sign=100%
CV>=7_q0.98_span1500: ρ=0.82, sign=100%
CV>=7_q0.985_span1000: ρ=0.81, sign=100%
CV>=7_q0.985_span2000: ρ=0.94, sign=100%
log2FC, bottom half (higher depth)
log2FC, top half (lower depth)